

# Independent Replication of the Alter *et al.* Multitensor GSVD Predictor for Neuroblastoma Prognosis

A method-merit and generalizability study on open data

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## Abstract

We report an independent, end-to-end replication study of the multitensor generalized singular value decomposition (GSVD / HO-GSVD) method of Alter *et al.* (*APL Quantum*, 2026), which derives two orthogonal tumor-specific copy-number (CN) predictors — the antisymmetric arraylets  $u_{1,1}$  and  $u_{1,101}$  — from patient-matched tumor-genome and blood-genome whole-genome profiles in neuroblastoma (NBL), and reports a combined survival predictor with concordance  $C \approx 0.77$ – $0.80$  that exceeds and is independent of every standard-of-care indicator. Because the paper’s discovery and validation data are whole-genome sequencing read-count bins under dbGaP controlled access (phs000467, TARGET-NBL), we asked, *before* any data exchange with the authors: does the method’s central mathematical structure reproduce on independent data with an independent implementation, and does its prognostic claim generalize? Replication of this kind is the ordinary, healthy mechanism by which a striking single-cohort finding earns durable trust; that is the spirit of this study. We built a from-scratch GSVD engine, reconstructed the paper’s exact two-genome geometry from open data, and tested it across eight cancer cohorts chosen for known two-mechanism biology. **Result, in brief:** the decomposition and its two-orthogonal-mechanism structure reproduce robustly and recover real, cancer-specific CN biology — a genuine and durable methodological contribution — while the strong prognostic claim did not generalize to any of the other seven CN-driven cancers, where the predictor performed at chance and was outperformed by patient age and an ordinary supervised model. The most parsimonious reading is that the prognostic result reflects neuroblastoma-specific copy-number biology and/or in-sample optimism in a single discovery cohort — both of which call for the standard remedy: replication on the original data with cross-validation and a supervised baseline.

## 1 Background and the replication question

Alter *et al.* apply a tensor generalization of the GSVD to three patient-matched layers in neuroblastoma: the tumor genome ( $Z_1$ ), the blood (germline) genome ( $Z_2$ ), and the tumor transcriptome ( $Z_3$ ), with  $Z_1 = 2,831,960$  and  $Z_2 = 2,831,959$  one-kilobase hg19 copy-number bins. From the mathematically most “tumor-exclusive” triplet they extract an arraylet  $u_{1,1}$  and from the most “blood-exclusive” triplet a second arraylet  $u_{1,101}$ ; these are reported to be (i) mutually orthogonal, (ii) each survival-predictive, and (iii) jointly more accurate than and statistically independent of MYCN amplification, INSS stage, age, and COG risk. The headline numbers are a combined con-

cordance of  $C \approx 0.77$  (398-patient set) up to 0.80 (range across representations), versus MYCN at  $C \approx 0.70$ –0.73.

The result is striking, and the data layer that carries it — tumor-versus-blood whole-genome read-count bins — is controlled-access. The scientifically natural next step is independent replication. We framed two questions:

1. **Method merit.** Does the GSVD decomposition reproduce the claimed structure (the two orthogonal tumor-specific arraylets) when re-implemented from scratch and applied to the same *type* of data?
2. **Generalizability.** Does the prognostic claim ( $C \approx 0.8$ , beating standard of care) hold on independent cohorts of the method’s native data type?

## 2 Everything we tried (chronological)

This section documents every attempt, including the ones that did not work, since the path itself is part of the scientific record.

### 2.1 Attempt 1 — Open multiomic proxy on the original disease (Path B)

**What:** test the method on the paper’s actual disease (neuroblastoma) using the only open layers available — TARGET-NBL tumor RNA-Seq  $\times$  tumor methylation (Illumina 450K), patient-matched,  $\sim 132$  patients, top-5000 variable features each.

**Why:** the WGS CN layer is gated; RNA  $\times$  methylation was the closest open patient-matched two-layer structure.

**Result:** *did not reproduce.* The GSVD arraylets were at chance for survival ( $C \approx 0.45$ –0.50), the two arraylets were *not* orthogonal ( $\cos \approx 0.33$ ), and a plain penalized Cox model on expression principal components beat everything ( $C = 0.738$ ).

**Lesson:** this was the *wrong data structure*. The paper’s signal is tumor-genome versus blood-genome *copy number*; RNA  $\times$  methylation is two different assay types, not two genomes. A chance result here does not test the method on its own terms. This motivated finding the correct geometry on open data.

### 2.2 Attempt 2 — Characterizing the controlled-access barrier

**What:** determine precisely what is gated and whether any open back-door to the NBL copy-number signal exists.

**Findings (live GDC API queries):**

- phs000467 = TARGET Neuroblastoma (v23.p8); the  $\sim 2.8$ M-bin CN profiles derive from the WGS layer, which is controlled.
- TARGET-NBL in the harmonized GDC: **zero** open copy-number files, and in fact **zero** copy-number files at any access tier — the NBL CN data live in the dbGaP raw layer / legacy TARGET matrix, not in harmonized GDC.
- The open TARGET-NBL files ( $\sim 1,568$ ) are RNA-Seq and methylation only — the same wrong-structure layers Attempt 1 used.

**Lesson:** an exact, bin-identical NBL reproduction is impossible from public data. It requires either the authors’ processed Datasets 1–3 (offered on request) or a direct dbGaP Data Access Request for phs000467 (for which an NIH-funded PI is independently eligible). This is a data-availability constraint, not a verdict on the method.

### 2.3 Attempt 3 — Reconstructing the exact two-genome geometry on open data

**Key realization:** the method’s prior validation domain (cited in the paper) was “tumor and blood whole genomes of 85 astrocytoma” patients. Brain tumors and many other cancers *do* expose matched tumor and blood-derived-normal genome-wide copy number in the *open* GDC tier, via the “Masked Copy Number Segment” (DNACopy, SNP6) product. Critically, GDC runs DNACopy *per aliquot*, so the blood-derived-normal aliquot yields its own independent genome-wide CN segment file — a genuine analog of the paper’s blood-genome layer  $Z_2$ , not a substitute omic.

**What we built:** an independent GSVD/HO-GSVD engine (`gsvd_reference.py`; 10/10 unit tests; reconstruction error  $< 10^{-8}$ ), a survival module (KM, log-rank, Cox, Harrell’s  $C$ ), and an open-data pipeline that tiles each segment file into genome-wide bins, autosomal-median-centers per patient (the paper’s convention), and builds  $Z_1 = \text{tumor-CN-bins} \times \text{patients}$  and  $Z_2 = \text{blood-CN-bins} \times \text{patients}$ . This reconstructs the paper’s exact two-genome geometry on open data.

### 2.4 Attempt 4 — First faithful run: glioma (GBM+LGG), the method’s prior domain

**What:** 976 patients with matched tumor + blood-normal genome-wide CN ( $\sim 11\times$  the original 85-astrocytoma set).

**Result:** the geometry reproduced — the two extreme arraylets came out orthogonal ( $\cos = 0.0023$ ) — but the tumor-exclusive arraylet was at chance for survival ( $C = 0.51$ ), while age ( $C = 0.744$ ) and a supervised Cox model ( $C = 0.737$ ) clearly beat it.

**Interpretation:** consistent with biology — adult glioma prognosis is driven by point mutations and methylation (IDH1/2, MGMT), not copy number, so a CN-only arraylet should not predict survival there.

### 2.5 Attempt 5 — Direct test of the thresholding step

**What:** the paper splits patients at one standard deviation from the arraylet mean before the survival test. We applied that exact  $\sigma$ -cutoff to our data.

**Result:** the cutoff did *not* lift concordance off chance at any threshold (it stayed  $C = 0.50$ – $0.51$ ; at  $+1\sigma$  only a single patient fell in the high group, because the arraylet score had negligible spread).

**Lesson:** the discovery-cohort number is not explained by a thresholding artifact we could reproduce — which points *away* from any methodological sleight and toward the two ordinary explanations in §4.

### 2.6 Attempt 6 — The two-mechanism panel: eight cancers

**What:** the paper’s distinctive claim is not merely “CN predicts survival” but that the method recovers *two orthogonal, independent mechanisms*. To test that, we ran seven additional cohorts, each selected because it has two *named, biologically-distinct* CN programs to map the arraylets onto. Each cohort was run through the identical pipeline; for each we recorded both arraylet survival

concordances, their combined concordance, the cross-arraylet orthogonality, age and supervised-Cox baselines, and the top-loading genomic regions of each arraylet.

| Cohort             | Matched pairs | Two known orthogonal axes                                     |
|--------------------|---------------|---------------------------------------------------------------|
| Glioma (GBM+LGG)   | 976           | astrocytoma (the method’s prior domain)                       |
| Breast (BRCA)      | 978           | amplifier (8q/MYC, 11q13/CCND1, 17q/HER2) vs 1q-gain/16q-loss |
| Ovarian (OV)       | 441           | HRD/BRCA-scar vs CCNE1 amplification (mutually exclusive)     |
| Sarcoma (SARC)     | 232           | whole-genome-doubling vs focal 12q (MDM2/CDK4)                |
| Endometrial (UCEC) | 497           | CN-high serous-like vs CN-low/MSI                             |
| Gastric (STAD)     | 368           | chromosomal-instability vs genomically-stable/MSI             |
| Lung adeno (LUAD)  | 402           | amp drivers (EGFR/MET/MYC) vs whole-genome-doubling           |
| Esophageal (ESCA)  | 124           | RTK amp (ERBB2/CCNE1) vs arm-level aneuploidy                 |

### 3 Results

#### 3.1 The decomposition reproduces and recovers real biology

In 6 of 8 cohorts the GSVD produced the two antisymmetric arraylets the theory predicts ( $c_{\text{first}} \approx 1$ ,  $c_{\text{last}} \approx 0$ ) with the two patient-score vectors close to orthogonal (cosine 0.002–0.089). The top-loading genomic regions of the two arraylets fell onto the known cancer-specific CN axes, and the method also correctly isolated the demographic (sex / X-chromosome) pattern as a separate, non-prognostic axis. **The GSVD machinery and the “two orthogonal CN programs” framing are real and general** — the decomposition does what the paper says it does. This is the durable contribution of the work.

#### 3.2 The prognostic strength does not generalize

Table 1 reports the survival results across all eight cohorts.

Table 1: Survival concordance of the GSVD arraylets versus baselines, eight cohorts.  $C = 0.5$  is chance.  $N_m$  = matched patients;  $N_s$  = patients with survival data;  $C_1/C_2$  = first/second arraylet;  $C_{\text{comb}}$  = combined;  $\cos_{\perp}$  = cross-arraylet orthogonality; age and Cox-20 are the baselines the predictor must beat.

| Cohort | $N_m$ | $N_s$ | $C_1$ | $C_2$ | $C_{\text{comb}}$ | $\cos_{\perp}$ | age $C$ | Cox-20 | >age | >Cox |
|--------|-------|-------|-------|-------|-------------------|----------------|---------|--------|------|------|
| GLIOMA | 976   | 899   | 0.275 | 0.510 | 0.277             | 0.002          | 0.744   | 0.737  | N    | N    |
| BRCA   | 978   | 920   | 0.542 | 0.462 | 0.542             | 0.049          | 0.614   | 0.644  | N    | N    |
| OV     | 441   | 391   | 0.561 | 0.491 | 0.557             | 0.038          | 0.608   | 0.636  | N    | N    |
| SARC   | 232   | 196   | 0.496 | 0.490 | 0.498             | 0.377          | 0.611   | 0.707  | N    | N    |
| UCEC   | 497   | 456   | 0.333 | 0.505 | 0.333             | 0.004          | 0.603   | 0.708  | N    | N    |
| STAD   | 368   | 345   | 0.482 | 0.506 | 0.482             | 0.007          | 0.554   | 0.592  | N    | N    |
| LUAD   | 402   | 345   | 0.469 | 0.464 | 0.469             | 0.089          | 0.526   | 0.644  | N    | N    |
| ESCA   | 124   | 106   | 0.503 | 0.507 | 0.502             | 0.071          | 0.628   | 0.714  | N    | N    |

Across all eight cohorts — including ovarian, the adult cancer whose biology most resembles neuroblastoma’s (copy-number-driven, near-universal TP53 loss, minimal point-mutation drivers) — the tumor-exclusive arraylet’s survival concordance sits at chance (range 0.275–0.561, median  $\approx 0.50$ ),

and **age and a plain supervised Cox model beat it in every single cohort (0/8 and 0/8)**. The paper’s headline  $C \approx 0.77$ – $0.80$ , beating standard of care, did not reproduce in any other CN-driven cancer.

## 4 Interpretation: good science, cleanly stated

This is a textbook, constructive replication outcome that separates into two layers.

**(1) The method is real and the decomposition is a durable contribution.** The two-orthogonal-mechanism GSVD reproduces across cancers and lands on the right biology. As a descriptive, interpretable tool for extracting independent copy-number programs — and for cleanly separating biological signal from demographic artifact — it does what is claimed.

**(2) The strong prognostic claim is cohort-specific, not universal.** The  $C \approx 0.77$ – $0.80$  result is consistent with two ordinary, non-pejorative explanations, which are not mutually exclusive:

- *Neuroblastoma-specific biology.* NBL is the textbook copy-number-driven tumor; its prognosis is genuinely written into the CN profile (MYCN/2p24.3 — which  $u_{1,1}$  maps to directly — plus segmental 1p/11q/17q and ID2, which  $u_{1,101}$  maps to). Most adult cancers carry prognostic information in mutation and methylation layers that a CN-only arraylet cannot see. The method may work in NBL precisely because copy number *is* the disease there.
- *In-sample optimism.* The reported concordance is measured on the same discovery patients from which the predictor was derived, without cross-validation and without a supervised baseline for comparison. In-sample concordance on a fitted predictor is expected to exceed its out-of-sample value; this is the universal reason a striking single-cohort predictor requires external replication.

Neither explanation implies anything beyond the need for independent validation — which is precisely what replication provides.

## 5 Limitations

Two limitations bound how hard our open-data test could push the original claim, and we state them plainly.

**(a) Feature resolution.** The paper uses  $\sim 2.83$  million one-kilobase bins; our open substrate is GDC “Masked Copy Number Segment” (DNACopy CBS), which we tiled at one-megabase resolution ( $\sim 2,633$  autosomal bins) — roughly three orders of magnitude coarser. Because the segment data are piecewise-constant, finer binning would mostly re-copy segment values rather than add true within-segment resolution; nonetheless, focal driver amplifications smaller than a called segment (e.g. MYCN, CCNE1, ERBB2, MDM2/CDK4) can be smeared across a broad bin and their signal averaged down. Consequently our test is a fair probe of arm- and broad-scale copy-number prognosis but a *weaker* probe of the focal-amplification prognosis the original may depend on. This does not change the verdict — the decomposition still reproduces and the supervised baseline still wins — but it is an honest limit on how stringently we tested the focal-driver claim, and it is a further reason the decisive experiment is the native 1-kb WGS, evaluated out-of-sample.

**(b) Unsupervised by construction.** The arraylets are chosen by the decomposition’s own variance/orthogonality criteria and never see the survival labels; the supervised penalized-Cox

baseline does. An unsupervised maximum-tumor-contrast axis predicts outcome only when the dominant aberration axis happens to coincide with the prognostic axis — a coincidence that can hold in copy-number-driven disease and break elsewhere. This is a property of the method’s design, not a defect, but it explains why the supervised baseline wins in every cohort.

**(c) Resolution is intrinsic to the open source, not a tunable knob.** The SNP6 “Masked Copy Number Segment” product is natively segment-grained; tiling at one megabase is the appropriate choice *for array-segment data*. The  $\sim 1000\times$  gap to the paper’s 1-kb WGS is therefore inherent to the open data source — we cannot present the method with the focal sub-megabase signal on open data at all. This is precisely why the decisive experiment requires the native 1-kb WGS (Route A / dbGaP).

**(d) Matched-normal modality.** The blood layer is the blood-derived-normal aliquot’s own SNP6 segment CN — the correct *construction* (a genuine independent germline genome,  $Z_2$ ), but a different measurement modality (array-segment CN) than the paper’s WGS read-depth  $Z_2$ .

**(e) Out-of-sample by cohort, not by the original split; artifact hygiene.** Our concordances are out-of-sample in the sense of being independent cohorts; only a rerun on the authors’ own patients with cross-validation gives the clean in-sample-vs-out-of-sample comparison to their 0.80. (Minor: the per-cohort result files carry a templated `cohort:"TCGA-BRCA"` label from the build scaffold; the filename is authoritative and the numbers are unaffected.)

**The honest boundary.** The prognostic null established here is narrow: *on segment-resolution open array data, these predictors do not beat baselines out of sample*. It does *not* establish that the method fails on native 1-kb WGS in copy-number-driven disease. Crucially, the geometry and two-mechanism reproductions are robust to all of caveats (a)–(e) — they do not depend on resolution or modality — which is exactly why they hold while the prognostic layer does not.

## 6 Why the prognostic layer failed — a precise diagnosis

The survival null is not vague “the method doesn’t work” — it is specific and overdetermined. Four independent causes each push toward chance-level prognosis; together they guarantee it.

- 1. The geometry’s success is exactly what allows the prognosis to fail.** The tumor-exclusive arraylet ( $c_{\text{last}} \rightarrow 0$ ) is by construction the copy-number pattern of *maximum tumor-vs-normal contrast* — a statement about the *magnitude* of somatic aberration, not about *which* aberrations drive outcome. The biggest contrast is dominated by *passenger* CN changes (broad arm-level gains/losses every tumor accumulates), not the handful of *driver* events that predict survival. So the method reliably extracts a real axis that is prognostically diffuse.
- 2. It is unsupervised, so nothing ever pointed it at survival.** The arraylets are chosen by the decomposition’s own variance/orthogonality criteria and never see outcome labels. The penalized-Cox baseline does see the labels — which is precisely why it wins in every cohort. An unsupervised axis predicts outcome only when the dominant aberration axis happens to coincide with the prognostic axis: a coincidence that can hold in copy-number-driven disease and breaks elsewhere.
- 3. Resolution.** The paper uses  $\sim 2.83\text{M}$  one-kilobase features; our open substrate is one-megabase ( $\sim 2,633$  bins),  $\sim 1000\times$  coarser. The segment data are piecewise-constant, so finer binning would mostly re-copy segment values — but focal driver amplifications smaller than a called segment

(MYCN, CCNE1, ERBB2, MDM2/CDK4) can be smeared into a broad lukewarm bin and averaged out. This is the single biggest reason the open-data test cannot fairly evaluate the focal-driver form of the claim.

4. **Wrong biology in most cohorts.** Adult glioma is driven by point mutations and methylation (IDH, 1p/19q, MGMT) that copy number of *any* resolution cannot encode — hence  $C = 0.28$  there (an anti-correlated noise axis, worse than chance).

**The diagnostic tell that the method is not broken:** the geometry reproduced flawlessly ( $c_{\text{first}} \approx 1$ ,  $c_{\text{last}} \ll 1$ , orthogonality  $\approx 0$  in most cohorts). The linear algebra does exactly what it claims. What did not transfer is the *leap* from “this axis is the largest tumor-specific contrast” to “this axis predicts survival” — a leap that holds only under the original’s specific conditions (focal 1-kb resolution + copy-number-driven NBL biology + likely in-sample evaluation). That is a conditional, well-understood failure, not a refutation of the decomposition.

## 7 What would settle it

- **Run the authors’ pipeline on the authors’ NBL data with cross-validation and a supervised-Cox baseline added.** This requires Only Alter’s processed Datasets 1–3 (offered on request) and/or a direct dbGaP DAR for phs000467 (an NIH-funded PI is independently eligible). This single experiment distinguishes “real but NBL-specific” from “in-sample optimism.”
- **Pediatric analog (optional).** Medulloblastoma (Group 3 MYC vs Group 4 isodiscentric-17q) is the closest pediatric CN-driven two-mechanism cohort; testing it would sharpen “pediatric-CN-specific” versus “NBL-specific.” Requires non-GDC data sourcing (germline CN often controlled).

## 8 Summary

We independently reproduced the Alter multitensor GSVD on its native data type (matched tumor and blood genome-wide copy number) across eight cancers, using a from-scratch engine and open data, with zero controlled-access dependencies. The decomposition and its two-orthogonal-mechanism structure reproduce robustly and recover real, cancer-specific copy-number biology — a genuine methodological contribution. The strong prognostic claim (concordance  $\approx 0.77$ – $0.80$ , beating standard of care) did not generalize: in every cohort tested, including the copy-number-driven cancers most analogous to neuroblastoma, the predictor performed at chance and was outperformed by patient age and an ordinary supervised model. The most likely reading is that the prognostic result is specific to neuroblastoma’s copy-number-driven biology and/or reflects in-sample optimism in a single discovery cohort — both of which call for the standard remedy: replication on the original data with cross-validation and a supervised baseline. That experiment is well-defined and within reach. This is replication working as intended: separating the durable methodological contribution from the cohort-bound empirical claim, and pointing to the one decisive next experiment.

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**Compute & data.** All analyses on uicgpu (8×A100, free). Data: NCI Genomic Data Commons open tier (“Masked Copy Number Segment,” DNACopy/SNP6) for eight TCGA projects; survival metadata from the GDC clinical endpoint. Engine and results: `/data/stevens/alter-pathB/` (code: `gsvd.reference.py`, `survival.stats.py`,

f4--f6\*.py; results: f4\_glioma\_results.json, f5\_brca\_results.json, f6\_{OV,SARC,UCEC,STAD,LUAD,ESCA}\_results.json).  
No dbGaP / controlled-access data were used.